

Propensity score matching with clustered data. An application to the estimation of the impact of caesarean section on the Apgar score

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This article focuses on the implementation of propensity score matching for clustered data. Different approaches to reduce bias due to cluster-level confounders are considered and compared using Monte Carlo simulations. We investigated methods that exploit the clustered structure of the data in two ways: in the estimation of the propensity score model (through the inclusion of fixed or random effects) or in the implementation of the matching algorithm. In addition to a pure within-cluster matching, we also assessed the performance of a new approach, 'preferential' within-cluster matching. This approach first searches for control units to be matched to treated units within the same cluster. If matching is not possible within-cluster, then the algorithm searches in other clusters. All considered approaches successfully reduced the bias due to the omission of a cluster-level confounder. The preferential within-cluster matching approach, combining the advantages of within-cluster and between-cluster matching, showed a relatively good performance both in the presence of big and small clusters, and it was often the best method. An important advantage of this approach is that it reduces the number of unmatched units as compared with a pure within-cluster matching. We applied these methods to the estimation of the effect of caesarean section on the Apgar score using birth register data. Copyright © 2016 John Wiley & Sons, Ltd.

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1. Introduction

In observational studies, direct comparison of outcomes across treatment groups can give rise with biased estimates because groups being compared may be different due to lack of randomization. Subjects with certain characteristics may have higher probabilities than others to be exposed to the treatment. If these characteristics are also related to the outcome under investigation, an unadjusted comparison of the groups is likely to produce wrong conclusions about the treatment effect.

Propensity scores, defined as the probability to receive the treatment conditional on the set of observed variables, were introduced by Rosenbaum and Rubin [1] as a one-dimensional summary of the multi-dimensional set of covariates, such that when the propensity scores are balanced across the treatment and control groups, the distribution of all covariates are balanced across the two groups. In this way, the problem of adjusting for a multivariate set of observed characteristics reduces to adjusting for the one-dimensional propensity score. (See Austin [2] for a review on the use of propensity score methods in the medical literature).

Importantly, propensity score methods can only ensure balance of background variables used in its estimation, and consequently, causal inferences based on these methods carry an assumption that no unobserved confounders exist. In medicine, as well as in many other fields, data often show a clustered structure where individual units are nested into clusters (e.g. patients nested into hospitals as in our motivating case study). In these cases, bias can arise from omission in the propensity score model of

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individual and/or cluster-level confounders. Multilevel models for the outcome are commonly applied to study datasets with a clustered structure [3, 4]. However, in the context of unstructured data, it has been shown that, when treatment groups differ substantially, regression adjustment for the estimation of causal effects heavily relies on model extrapolation and can give rise to biased estimates [5].

Methods based on the propensity score (such as matching, weighting and stratification, see [6, 7] for a review) are a more robust alternative, but only relatively recently has the treatment effect literature started to consider the implementation of propensity score methods with clustered data. Among the few works on the topic, Arpino and Mealli [8] showed, through an extensive simulation study, the benefit of using random-effects or fixed-effects models for the estimation of the propensity score to reduce the bias due to unmeasured cluster-level confounders when propensity scores are used to match treated and control individuals. Other studies have shown the benefits of exploiting the multilevel structure in the implementation of propensity score stratification [9] and weighting [10].

In this paper, we focus on propensity score matching (PSM) and consider different approaches to take into account the clustered structure of the data with the aim of reducing the bias due to cluster-level confounders. We consider methods that exploit the information on the clusters to which units belong in two ways: in the estimation of the propensity score model via the inclusion of fixed or random effects and in the implementation of the matching algorithm.

As for the latter approach, the simplest idea to adjust for cluster effects is to implement the matching algorithm within each cluster. In fact, if we impose that treated and matched control units must belong to the same cluster, we automatically achieve a perfect balance of all the observed and unobserved cluster-level characteristics. This approach was not considered in Arpino and Mealli [8] because their focus was on datasets characterized by very small-sized clusters, a case where within-cluster matching is not feasible.

When clusters sizes are big enough, within-cluster matching is a valid strategy, but it can still imply losing many units that cannot find a match because the search is forced to be within clusters [11]. Discarding unmatched units is problematic because it may imply a change of the estimand [12]. In addition to a pure within-cluster matching, we also propose and assess the performance of an approach that has not been tested in previous studies. This approach first searches for control units to be matched to treated units within the same cluster. If matching is not possible within-cluster, then the algorithm searches in other clusters. This approach, which we define 'preferential' within-cluster matching (PW), is expected to carry the benefits of pure within-cluster matching (in terms of bias reduction) and matching on the pooled dataset (in terms of minimizing the number of unmatched units).

We evaluate the performance of the different approaches in the context of a real case study and using a simulation set-up that mimics the available data. As a motivating case study, we consider estimation of the effect of caesarian versus vaginal section on the probability of low Apgar score, a well-known measure of child's health. Our data are characterized by a strongly unbalanced structure with several hospitals with big sample sizes and some hospitals with small sample sizes. This type of data structure, which is not uncommon in real applications, has not been considered in previous Monte Carlo studies on propensity score techniques with clustered data. Moreover, our simulation experiments set the distribution of variables and effect sizes to mimic the distributions and associations observed in the empirical analysis. Thus, our approach, similarly to Huber *et al.* [13], is less prone to the standard critique of simulation studies that the chosen data generating processes are irrelevant for real applications. Given the lack of applications of propensity score methods with multilevel data, our paper serves also as an illustrative example on the usefulness of these methods in the context of a medical study.

In our motivating case study, the relevant structure is represented by a hierarchy of two levels (individuals nested into hospitals), and we will consider this type of data structure in the following. However, the approaches we consider can be easily adapted to more complex structures. We focus on treatments assigned at the individual level. For propensity score methods applied to cluster randomized trials, see Leyrat *et al.* [14, 15].

The rest of the article proceeds as follows. Section 2 introduces our motivating application and data. In Section 3, we discuss alternative PSM procedures for multilevel data. Section 4 presents a simulation study that mimics the data structure of the motivating example to assess the performance of the different methods under study. We also consider three variants of the baseline simulation study. In Section 5, we report the results from the application of the proposed methods to the motivating case study. Section 6 offers some concluding remarks.

2. Motivating case study

As motivating case study, we consider the estimation of the causal effect of caesarean section (treatment) versus vaginal delivery (control) on the 5-min Apgar score, a widely employed indicator of the clinical state of newborns [16, 17].

The 5-min Apgar score is a clinical test performed on a newborn 5 minutes after birth. It is a composite measure of breathing effort, heart rate, muscle tone, reflexes and skin colour [18]. Each item is scored 0, 1 or 2, and thus, the total score ranges from 0 to 10. Infants with a score higher than 6 are usually considered normal [19]. Low Apgar score is strongly associated with abnormal future development of the child [20] and infant mortality risk [21]. Following the literature, as outcome variable, we consider a binary variable indicating whether the Apgar score is below the threshold of seven points ('low' Apgar score) or not. In our sample, the proportion of low Apgar scores is 6.75%. This is an example of low-incidence binary outcome for which PSM has been found to perform considerably better than logistic regression adjustment on unstructured data [22].

The treatment considered in this study is the caesarean section, arguably one of the most common obstetric procedures. The intervention rate of caesarean delivery has grown substantially in the last decades [23]. Nonetheless, there is concern about whether the current high rates of caesarean sections are justified because the procedure is not without risk and previous studies reported negative effects on both the mother and the infant [24, 25]. In particular, some studies also found a negative effects of caesarean sections on the Apgar score [26].

Similarly to several medical studies, in our empirical analyses, we use data collected in different sites (hospitals in our case). The dataset we consider contains information on deliveries occurred in the 22 hospitals of the Italian region of Sardinia in 2010 and 2011. The source is the official abstract of the birth event which is filled by physicians after each birth. This form is divided in three parts containing information on the mother, the pregnancy and the infant. Our dataset includes all hospitalized births in the period 2010–2011.

Following other observational studies on caesarean section (e.g. [23, 27]), to better isolate the effect of caesarean section on the target variable, we focus on the subset of non-complicated pregnancies. In particular, we selected nulliparous women at 32 or more weeks of gestational age with a singleton and living infant in vertex (head-down) position and without birth anomalies. We further restrict the sample to mothers aged between 15 and 44. These selection criteria are part of the matching strategy in the sense that they avoid including in the working sample women with extremely high probabilities of undergo a caesarean section and that could not find comparable control units. Our working sample (Table I) contains 14 757 observations clustered in 20 hospitals (two hospitals were removed because after applying the selection criteria they only contained treated or untreated women).

A simple unadjusted comparison of Apgar scores by mode of delivery indicates a negative effect of caesarean section. In fact, in our sample, the prevalence of low Apgar scores is 10.9% and 5.2% among treated and control women, respectively. Obviously, this unadjusted comparison ignores differences in individual and hospital characteristics between the groups of mothers that undergo caesarean or vaginal delivery. Indeed, previous studies have identified several clinical and socio-economic variables that proved to be associated with caesarean section and Apgar scores. Additional to individual-level confounders, the literature suggested the relevance of hospital-level factors both on the decision of taking a medical treatment and on medical outcomes. For example, Caceras *et al.* [27] and Bragg *et al.* [23] found that even after adjusting for socio-demographic and clinical factors, the rates of cesarean deliveries varied according to the hospital, suggesting the importance of hospital practices and culture in determining a hospital cesarean rate. Examples of relevant hospital-level variables include obstetricians practices, physicians preferences and guidelines promoting or restricting the liberal use of caesarean sections. Substantial variability in the outcomes of medical treatments across hospitals after controlling for clinical variables have been found due to differences in the quality of care [28, 29].

In our dataset, overall, 35% of births were caesarean, with a range of 11–64% across hospitals (Table I). Substantive variation across hospitals was also observed in the prevalence of low Apgar scores, which ranged from 0 to 16.5 cases every 1000 deliveries, with an average of 6.75. Apart from an uneven distribution of individual-level risk factors, these variations across hospitals may indicate the presence of hospital-level important factors. In general, it is difficult or impossible to measure all hospital characteristics that can impact on the probability to receive a treatment and on its outcomes, as they may be not available in official forms. Our case is no exception, and thus, standard PSM would fail to adjust

Table I. Distribution of births, caesarean sections and low Apgar scores by hospital.

Hospital	Number of births	Number of caesarean sections	Caesarean sections (%)	Low Apgar scores (%)
1	2532	1166	46.0	16.5
2	1788	623	34.8	2.7
3	1687	540	32.0	5.3
4	1473	632	42.9	14.2
5	1253	410	32.7	0.7
6	1197	428	35.7	3.3
7	980	240	24.4	2.0
8	875	238	27.2	5.7
9	529	190	35.9	3.7
10	434	135	31.1	6.9
11	403	164	40.6	0.0
12	396	117	29.5	7.5
13	351	134	38.1	8.5
14	266	74	27.8	7.5
15	208	99	47.5	9.6
16	191	122	63.8	10.4
17	103	40	38.8	9.7
18	50	9	18.0	20
19	32	13	40.6	0.0
20	9	1	11.1	0.0
Total	14 757	5375	35.0	6.75

for hospital-level omitted variables. In the next section, we discuss strategies that exploit the information on the hospital where each mother has delivered to adjust for unobserved hospital-level confounders.

3. Propensity score matching with clustered data

Consider a two-level data structure where N individual-level units, indexed by i ($i = 1, 2, \dots, n_j$), are nested in J second-level units (clusters), indexed by j ($j = 1, 2, \dots, J$). We consider a binary treatment administered at the individual level, T , and an outcome variable, Y , also measured at the individual level. Confounders can be first (X) or second-level (Z) variables.

Under the potential outcome framework, let $Y_{ij}(t)$ be the potential outcome if unit ij was assigned to treatment t , $t \in \{0, 1\}$. The fact that potential outcomes are labelled only by ij and t corresponds to the Stable Unit Treatment Value Assumption [30]. This usually invoked assumption requires that potential outcomes for a unit are not affected by the treatment received by other units, and that there are no hidden versions of the treatment. Under Stable Unit Treatment Value Assumption, for each unit ij , an individual causal effect is defined as a comparison of $Y_{ij}(1)$ with $Y_{ij}(0)$, yet only one of the two potential outcomes is observed depending on the value taken by T_{ij} .

Usually, the Average Treatment effect on the Treated (ATT) is considered as an interesting summary of individual causal effects: $ATT = E(Y_{ij}(1) - Y_{ij}(0) | T_{ij} = 1)$. To identify the ATT with observational data, the following assumptions are often invoked:

- Unconfoundedness: $Y(1), Y(0) \perp T | (X, Z)$;
- Overlap: $0 < P(T = 1 | (X, Z)) < 1$.

Unconfoundedness asserts that the probability of assignment to a treatment does not depend on the potential outcomes conditional on observed covariates [1]. Unconfoundedness essentially assumes that within subpopulations defined by the values of the covariates, we have random assignment of the treatment. This assumption is often referred to also as selection on observables because it rules out the role of unobserved variables [6]. Overlap implies that, for all possible values of the covariates, there is a positive probability of receiving each treatment. Usually, analyses are restricted to the common support of covariates across treatment groups, where this assumption is met.

Under the previous assumptions, adjustment on the propensity score is sufficient to eliminate bias due to observed confounders [1]. The propensity score, e , is defined for each unit as the probability to receive the treatment given its covariate values: $e_{ij} = P(T_{ij} = 1 | (X_{ij}, Z_j))$. Rosenbaum and Rubin [1] proved

that the propensity score is a balancing score, that is, $(X, Z) \perp T | e(X, Z)$, meaning that at each value of the propensity score the distribution of the covariates defining the propensity score is the same in the treated and control groups. They also showed that if unconfoundedness holds conditional on covariates it also holds conditional on the propensity score, that is, $Y(1), Y(0) \perp T | e(X, Z)$. These results justify adjustment on the propensity score instead of on the full multivariate set of covariates.

Usually, in observational studies, the propensity score is not known and must be estimated from the data. Parametric models, such as logit or probit models, with inclusion of interactions and higher-order terms are commonly used [31]. An incorrectly estimated propensity score may fail to respect the balancing property [5]. Our focus is not on misspecifications of the functional form of the propensity score model but on the bias caused by omitted cluster-level confounders. If one or more variables affecting the selection into treatment and potential outcomes are not observed, then unconfoundedness is violated, and *ATT* estimators based on the propensity score will be biased. In the following, we shall assume that we have good measurement on all individual-level confounders, X , but we have no information on cluster-level confounders. For simplicity, we consider that only one cluster-level confounder exists, Z , which is unobserved. The methods we discuss in the following can be adapted when some cluster-level variables are observed and others are not.

Among propensity score methods available to adjust for an unbalanced distribution of covariates between treated and control groups, we consider propensity score matching (PSM). In particular, we consider one-to-one nearest neighbour matching within a maximum distance (caliper) of 0.20 standard deviations of the estimated propensity score [32]. For each treated unit in the sample, the algorithm searches for the closest control unit in terms of propensity score. If no control unit is available in the range defined by the caliper, the treated unit is discarded from the working sample. We considered matching with replacement, where the same control unit can be used several times as a match. Matching with replacement is expected to improve the quality of matches and therefore to reduce bias [33]. However, a bias-variance trade-off emerges because matching with replacement increases the variance of estimates [34]. Because our main focus is on the bias of the estimators, we considered matching with replacement. A drawback of matching with replacement is that the matched dataset can contain several replicates of the same control units implying that observations will not be independent. In the simulations, we calculated the number of times each control unit is used as a match and averaged this number over the set of control units that were used at least once. As we discuss in the next section, the number of control replicates is quite small, and therefore, the issue of non-independent observations is not particularly worrying in our analyses.

When the dataset has a two-level structure, one can consider different ways of implementing PSM. We compare the following methods:

- (1) Single-level propensity score model; matching on the pooled dataset ('Naive', NV);
- (2) Single-level propensity score model; within-cluster matching ('Within-cluster', W);
- (3) Single-level propensity score model; preferential within-cluster matching ('Preferential', PW);
- (4) Fixed-effects propensity score model; matching on the pooled dataset ('Fixed-effects', FE);
- (5) Random-effects propensity score model; matching on the pooled dataset ('Random-effects', RE).

The first approach, which we label as NV, uses a single-level logit model to estimate the propensity score:

$$\text{logit}(e_{ij}) = \alpha_0 + X_{ij}\beta. \quad (1)$$

Then, a one-to-one caliper matching on the estimated propensity scores is implemented on the pooled dataset. Formally, let I_1 and I_0 denote the set of treated and control units, respectively, and let indicate with A_{rj} the set of control units matched to $rj \in I_1$ (note that ij indicates a generic unit in cluster j while rj indicates a generic treated unit in cluster j):

$$A_{rj} = \left\{ kj' \in I_0 : \hat{e}_{kj'} = \min_{kj' \in I_0} |\hat{e}_{rj} - \hat{e}_{kj'}| < 0.20\hat{\sigma}_e \right\}. \quad (2)$$

After (2) has been constructed for all units in the treatment group, the matched dataset M is built so that all treated units that successfully found a matched control unit and all matched control units are included in M while all other units are discarded:

$$M = \{rj : A_{rj} \neq \emptyset\} \cup \left\{ \bigcup_{rj} A_{rj} \right\}. \quad (3)$$

The *ATT* estimator takes the form:

$$\widehat{ATT} = \frac{1}{card(M)} \left\{ \sum_{rj \in I_1 \cap M} \left(Y_{rj} - \sum_{kj' \in I_0} Y_{kj'} w(rj, kj') \right) \right\}, \quad (4)$$

where $w(rj, kj')$ is the weight assigned to the control unit kj' in the estimation of the unobserved potential outcome, $Y(0)$, for the treated unit rj . In principle, in a nearest neighbour matching, at most one control unit is matched to each treated unit. However, it is possible that several control units have the same value of the propensity score within the caliper ('ties'). In this case, A_{rj} will contain several units, and each control unit in A_{rj} enters the (4) with a weight equal to $1/card(A_{rj})$. Control units that are not matched to rj are assigned a weight of zero in (4).

Model (1) is a simple single-level logit model that ignores the multilevel data structure and in fact the subscript ij is used only for notational consistency with the following methods. The multilevel structure is also ignored in the implementation of the matching (2). In fact in the creation of the matched pairs, the cluster membership is ignored; there is no constrain on j and j' to be equal, while they can happen to coincide though.

Therefore, the naive approach ignores completely the hierarchical structure both in the estimation of the propensity score and in the implementation of the matching. This approach can succeed in balancing individual-level variables but is not meant to build a balanced matched dataset also with respect to cluster-level variables. In this case, if we do not include all relevant confounders at the cluster-level in the propensity score and obtain a good balance of all of them, an *ATT* estimator based on the naive approach will be biased.

The second approach, within-cluster matching (W), deals with this problem by matching units within clusters. In this case, the model used for the estimation of propensity scores is still model (1), and thus, it ignores the clustering. However, the matching is implemented within clusters in the sense that the algorithm applies the procedure (2) only to units belonging to the same cluster, that is, forcing $j = j'$. Therefore,

$$A_{rj} = \left\{ kj \in I_0 : \hat{e}_{kj} = \min_{kj' \in I_0} |\hat{e}_{rj} - \hat{e}_{kj}| < 0.20\hat{\sigma}_e \right\}. \quad (5)$$

As with the naive approach, also with the within-cluster approach, the matched dataset is created according to (3), and *ATT* is estimated using (4). Within-cluster matching automatically guarantees that all cluster-level variables (measured and unmeasured) are perfectly balanced. This can come to a cost. Control units to be matched with treated units are only searched within the same cluster. Therefore, it could be that we do not find a control matched unit that we would find in other clusters. So, we may have many more unmatched treated units than with approach A.

To avoid these problems and combine the benefits of the naive and within-cluster approaches, the third approach, PW, starts by searching control units within clusters (according to (5)). If none is found, control units are searched in other clusters (according to (2)). This approach improves the balancing of cluster-level variables with respect to the naive approach and reduces the number of unmatched treated units with respect to the within-cluster approach.

In alternative to exploiting the hierarchical structure in the implementation of the matching, the last two approaches take it into account when modelling the propensity score. In particular, approach RE and FE use random or fixed effects, respectively, to represent unmeasured cluster-level variables. A random-effects logit model can be represented as follows:

$$logit(e_{ij}) = \alpha_j + X_{ij}\beta, \quad (6)$$

where $\alpha_j \sim N(\alpha_0, \sigma_\alpha^2)$. To estimate propensity scores from (6), predictions of cluster effects α_j are plugged in together with estimates of covariates coefficients.

A fixed-effects logit model differs from (6) for the fact that α_j are not represented by realizations of a random variable but they are fixed intercepts for each cluster that can be estimated by including a set of binary indicators for $J - 1$ clusters. After propensity scores have been estimated, both approaches RE and FE implement the matching on the pooled dataset (procedure (2)).

Arpino and Mealli [8] showed that PSM using fixed-effects or random-effects models gave similar results in terms of bias reduction and mean squared error of *ATT* estimators in the presence

of unmeasured cluster-level variables and small-size clusters. We considered these approaches also in the case of a strongly unbalanced dataset where some clusters are big and others are small. We compared these approaches with the naive method and with the pure and preferential within-cluster matching (W and PW).

4. Simulation studies

In this section, we describe our simulation experiments aimed at comparing the performance of the different matching strategies described earlier in the presence of unobserved confounders at the cluster-level.

4.1. Set-up

We designed our simulation experiments to mimic the observed data in several respects. First, we kept the same data structure observed in our dataset, that is, the same number of clusters (hospitals) and the same clusters' sample sizes (Table II). In this way, in our simulations, we considered a realistic case

Table II. Simulation results after propensity score matching with replacement when the cluster-level confounder Z is unobserved. $ATT \approx 10.5\%$.

Metrics		Strategy [†]					
		Crude	NV	W	PW	FE	RE
$\beta_Z = 0$	Number of unmatched units	—	0.58	51.71	0.58	0.35	0.31
	Average number of controls replications	—	0.07	0.14	0.14	0.09	0.09
$ICC(Y) = 7.1\%$	ASB Z	4.97	5.07	0.00	0.29	0.93	0.93
$ICC(T) = 1.1\%$	ASB X	13.22	0.98	1.68	1.67	0.93	0.94
	ASB All	12.76	1.20	1.59	1.60	0.93	0.94
	% Bias	54.15	0.88	2.85	0.56	9.93	9.10
	$RMSE_{montecarlo}$	6.01	3.43	3.37	3.47	3.38	3.23
	SE	2.11	3.52	3.37	3.475	3.36	3.37
$\beta_Z = 0.2$	Number of unmatched units	—	0.65	52.98	0.65	0.71	0.71
	Average number of controls replications	—	0.08	0.16	0.16	0.10	0.10
$ICC(Y) = 7.5\%$	ASB Z	17.61	18.20	0.00	0.25	0.92	1.01
$ICC(T) = 2.2\%$	ASB X	13.03	1.03	1.70	1.69	0.95	0.96
	ASB All	13.29	1.98	1.61	1.61	0.95	0.96
	% Bias	59.24	11.29	3.06	1.50	10.05	10.37
	$RMSE_{montecarlo}$	6.63	3.55	3.52	3.62	3.42	3.41
	SE	2.15	3.63	3.46	3.42	3.44	3.42
$\beta_Z = 0.4$	Number of unmatched units	—	0.87	55.73	0.87	0.96	0.98
	Average number of controls replications	—	0.08	0.18	0.19	0.12	0.12
$ICC(Y) = 7.2\%$	ASB Z	35.36	35.89	0.00	0.37	0.86	0.94
$ICC(T) = 4.9\%$	ASB X	12.87	1.08	1.77	1.75	1.01	1.02
	ASB All	14.12	3.02	1.68	1.67	1.00	1.01
	% Bias	63.74	18.48	2.13	3.15	9.64	10.69
	$RMSE_{montecarlo}$	7.20	3.83	3.75	3.86	3.74	3.78
	SE	2.15	3.84	3.52	3.45	3.55	3.56
$\beta_Z = 0.6$	Number of unmatched units	—	0.84	61.31	0.84	0.66	0.71
	Average number of controls replications	—	0.09	0.23	0.23	0.16	0.16
$ICC(Y) = 6.7\%$	ASB Z	52.55	52.96	0.00	0.66	0.78	0.81
$ICC(T) = 9.0\%$	ASB X	12.72	1.17	1.99	1.95	1.10	1.09
	ASB All	14.93	4.05	1.88	1.88	1.09	1.08
	% Bias	67.70	25.97	0.07	6.34	9.89	10.26
	$RMSE_{montecarlo}$	7.69	4.40	3.68	3.92	3.90	3.97
	SE	2.15	3.98	3.53	3.36	3.72	3.66

[†]NV, naive; W, within-cluster; PW, preferential within-cluster; FE, fixed-effects; RE, random-effects.

ASB, absolute standardized bias; RMSE, root mean squared error; SE, standard error.

The proportion of matched pairs that are within clusters in PW is $\approx 99\%$.

with a strongly unbalanced structure where some clusters are big and others have small sample sizes (the cluster size varied from 9 to 2532 with an average of about 738 units per cluster). Second, instead of generating values of covariates as realizations of random variables as typically carried out in simulation studies, we used the same distribution of covariates as observed in the dataset. The only exception was for a cluster-level variable, Z , that we introduced to explore the confounding effect at the cluster (i.e. hospital) level. Finally, the coefficients of individual-level covariates in the true models generating the treatment and the outcome were set to values similar to observed coefficients estimated on the real data.

Given the complete set of covariates (X, Z) , the probability of being treated was generated according to the following:

$$e_{ij} = 1 / [1 + \exp(\beta_0 + \beta_1 X_{1ij} + \dots + \beta_k X_{kij} + \beta_Z Z_j + u_j)] \quad (7)$$

and the outcome was generated by the following model:

$$P(Y_{ij} = 1) = 1 / [1 + \exp(\gamma_0 + \gamma_1 X_{1ij} + \dots + \gamma_k X_{kij} + \gamma_Z Z_j + \alpha T_{ij} + v_j)], \quad (8)$$

where $\beta = [\beta_0, \dots, \beta_k, \beta_Z]$ and $\gamma = [\gamma_0, \dots, \gamma_k, \gamma_Z]$ are the vectors of coefficients, $X = [X_1, \dots, X_k]$ is the set of observed individual-level confounders and Z is the cluster-level confounder. The error terms u_j and v_j are realizations of normally distributed independent random variables with zero mean and standard deviation equal to 0.2. They represent random effects at the cluster-level. Values of Z were generated as realizations of a normal variable with $\mu_Z = 0$ and $\sigma_Z = 0.25$, which is the average standard deviation of the observed confounders. The coefficients β , γ and α in Equations (7) and (8) were set to values approximately equal to the corresponding coefficients estimated on the real data using logistic regression models without the cluster-level variable Z (see Section A of the Supporting Information for the exact values). This gave a true ATT equal to $\approx 10\%$ in all simulation experiments we considered. The variable Z was introduced in the true data generating models (7) and (8), but it was omitted in the implementation of the PSM strategies described in the previous section to represent the presence of an unobserved cluster-level confounder. We considered different levels of the unobserved confounder variable by fixing $\gamma_Z = 0.4$, while β_Z varied from a null to a strong confounding effect (as compared with the magnitude of the coefficients of the other covariates): $\beta_Z = \{0, 0.2, 0.4, 0.6\}$.

Figure 1 illustrates the true propensity score distribution in the group of treated and control units for $\beta_Z = \{0, 0.2, 0.4, 0.6\}$. More specifically, we considered one random replicate for each of the four values of β_Z . The figure shows that in all four cases there was a good overlap of the two distributions. This was also true when we varied some features of the baseline simulation study.

Under each scenario, 500 datasets were generated from models (7) and (8). For each simulated dataset, we employed the PSM methods described in the previous section to obtain a matched subset. The simulation experiments were implemented in R [35]. In particular, for the NV, FE and RE methods, we obtained the matched subsets using the function `Match` in the package `Matching` [36]. At the time of writing, neither this package nor others have an option for implementing within-cluster (W) and PW so we programmed a routine that makes use of the `Match` function (the code used in the baseline simulation is made available in Section B of the Supporting Information).

We summarized the results by averaging over the 500 replicates the following metrics calculated on each dataset: the number of unmatched treated units, the average number of times each control unit is used as a match (replications), the absolute standardized bias (ASB) of each confounder, the percent bias of the estimated ATT ($\%BIAS$), the root mean squared error ($RMSE_{montecarlo}$) and the estimated standard error of the ATT (SE). Note that matching with replacement (with a common caliper and a common propensity score model for all clusters) forces the relation: N . unmatched units (NV) = N . unmatched units (PW) $\leq$ N . unmatched units (W). The ASB was calculated for each confounder as the absolute value of the difference of means between treatment and control group standardized by its standard deviation in the treatment group. Similarly, for categorical variables, we created a dummy variable for each category and considered the standardized difference between the proportions of 1s observed in treated and control groups. The ASB is a measure of covariate balance: a lower ASB indicates that the treatment and control groups are more similar with respect to the given covariate. We report the average ASB for individual-level confounders (X), the ASB for the cluster-level

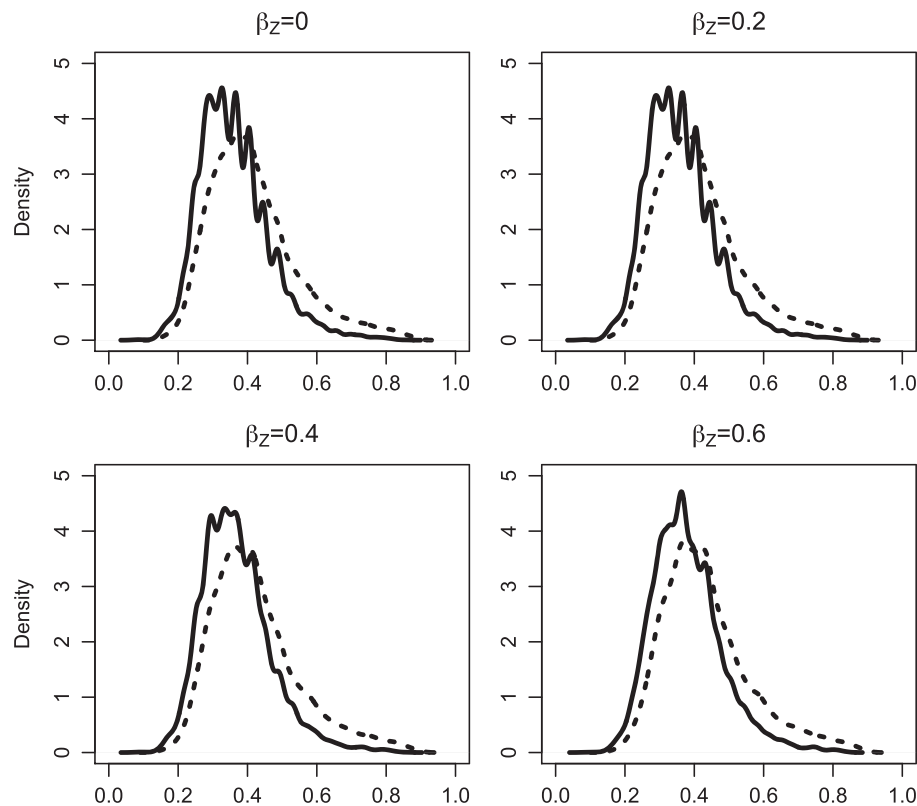


Figure 1. Density of the simulated propensity score distribution for treated (dotted line) and control units (continuous line).

confounder (Z) and for the average ASB of all confounders. The relative bias of each PSM estimators was calculated as

$$\left| \frac{\widehat{ATT} - ATT}{ATT} \right| \cdot 100, \quad (9)$$

where $\widehat{ATT}$ is the average treatment effect on the treated estimated using (4) and ATT is the true value of the average treatment effect on the treated.

The root mean squared error ($RMSE_{montecarlo}$) was calculated averaging the squared difference between the estimated and true ATT over the 500 replicates and taking the square root of it. We also calculated the standard error of the ATT (SE) using a linear regression model estimated on the matched dataset with clustered standard errors to take into account within-cluster dependence in the outcome. More specifically, we estimated a linear model where the outcome was Y and the only covariate was T and the standard error of ATT was calculated as the standard error of the coefficient of T . Bootstrapping is an alternative way to obtain standard errors. Apart from being computationally consuming, especially for a simulation study, bootstrapping is not valid for matching with a fixed number of matches [37]. Abadie and Imbens [38] provide consistent estimators of standard errors of matching-based estimators, but their use with multilevel data has not been considered yet.

In each table of simulations results, we also report the intra-class correlation coefficient (ICC) of the outcome, $ICC(Y)$, and of the treatment, $ICC(T)$. The ICC is a commonly used measure in multi-level analyses to quantify the variability of a variable across clusters and is defined as the ratio between the estimated variance of the cluster-level random effect and the sum of the individual-level and cluster-level random effects' variances [4]. In the tables' notes, for preferential matching, we also indicate the percentage of matched pairs that are within clusters.

To gain further understanding on the performance of the different considered PSM procedures, we expanded the simulation study in three ways:

- considering the case where Z is observed;
- varying the coefficients of the individual-level covariates across a subgroup of cluster;

- fragmenting the data set into a higher number of smaller clusters.

In the first case, the simulation design remained unchanged, but we modified the PSM to take into account the cluster-level variable. In other words, we considered the case where the researcher observes the cluster-level Z and uses it in estimating the propensity score. We expect that all methods benefit from observing Z . The only exceptions are the within-cluster matching and the fixed-effects approach. In the first case, all cluster-level variables are perfectly matched, while in the fixed-effects approach, cluster-level covariates cannot be used in the estimation of the propensity score because the whole cluster-level variability is absorbed by the fixed effects. Therefore, we modified only the propensity score models used in approaches NV, PW and RE to include the variable Z . Results for W and FE are reported from the previous analyses for comparison.

In the second variation of the baseline simulation set-up, the magnitude of the coefficients of the individual-level variables was increased by 25% in the true model generating the treatment (Equation 7) in a randomly chosen subset of 10 clusters (hospitals). All other conditions were kept as in the basic simulation set-up. This simulation study is motivated by the fact that in some hospitals some individual-level characteristics may be more important for the decision of using caesarean section than in others. Introducing stronger coefficients in some hospitals imply that the distribution of individual-level variables become more unbalanced in those hospitals between the groups of treated and control units. These varying coefficients by clusters could be modelled through the inclusion of random slopes (in the random-effects model of the RE approach) or interactions between individual-level covariates and hospital indicators (in the fixed-effects model of the FE approach). However, in the context of a practical application, it may be difficult to know a-priori which covariates should be allowed to have different slopes by clusters, and it may be computationally cumbersome to include many random slopes and cross-level interactions [39]. To assess the consequence of ignoring not only the unobserved cluster-level confounder but also the varying coefficients by clusters, we applied the five PSM methods without modifications. Within-cluster and preferential within-cluster PSM should partially deal with varying slopes across clusters because matching only or preferentially within clusters should improve the balance of those covariates having stronger effects on T within some clusters.

Finally, to assess how the performance of the five considered PSM procedures changes when clusters have smaller sample sizes, we considered four additional simulations fragmenting the data set into a higher number of smaller clusters. In practice, we fragmented the original clusters into smaller groups of size 50, 100, 200 and 300. Residual units were aggregated to form additional clusters. We kept constant all other characteristics of the data and of the baseline simulation set-up. In the presence of small clusters, within-cluster matching, which insists on matching within clusters only, is expected to perform poorly [10]. In this case, PW is expected to show a better relative performance.

4.2. Simulation results

Table II presents the results of the baseline simulation study introduced in the previous section. We considered four scenarios by varying the effect of the cluster-level unobserved confounder in the true treatment assignment model, $\beta_Z = \{0, 0.2, 0.4, 0.6\}$. For each scenario, we compare the performance of the five PSM strategies described in Section 3 in terms of unmatched units, balance (ASB), percent bias and root mean squared error ($RMSE_{montecarlo}$). We also report in the first column the results obtained without any adjustment ('Crude').

First of all, we notice that an unadjusted comparison between treated and control groups' outcomes gives strongly biased estimates (relative bias ranging approximately from 54% to 68%). All PSM methods guarantee a considerable reduction of the bias. However, as expected, only PSM methods adjusting for clustering (W, PW, FE and RE) achieve a low bias. The naive approach, which ignores clustering and only adjusts for imbalance in individual-level confounders, shows considerably higher relative biases, especially when the effect of the unobserved cluster-level confounder is the strongest (bias = 26%). The drawback of this method is highlighted by the ASB estimates. The naive method always yielded a very good balance (low ASB) for individual-level confounders, X , but was, of course, not able to improve the balance of the cluster-level variable, Z .

While the naive approach ignores completely the clustering, within-cluster matching forces matched pairs to belong to the same cluster. Within-cluster matching guarantees a perfect balance of the unobserved cluster-level variable ($ASB(Z) = 0$), at the expense of having only a slightly worse balance of individual-level variables as compared with the naive method (and also to the other methods). Similarly, preferential matching, which first tries to match units within-cluster and only when this is not possible

searches in other clusters, also attains a very good balance of both the cluster-level unobserved confounder and individual-level (observed) confounders. Within-cluster and preferential matching are the methods that reduce the bias the most: the relative bias for methods W and PW range between 0.07% and 6.34%, while for FE and RE methods, that use, respectively, a fixed-effects or random-effects logistic regression to estimate the propensity score, the relative bias is always around 9–10%, which can still be considered as acceptable.

Preferential matching performs particularly well when the effect of Z is low, while within-cluster matching performs better when the effect of Z is the strongest. Preferential matching has the advantage of reducing the number of unmatched treated units compared with within-cluster matching. The pure within-cluster matching, in fact, discards on average about 55 units (corresponding to about 1% of the treated units) as compared with less than 1 treated unit that, on average, remains unmatched with preferential matching. This corresponds to a percentage of within-cluster matched pairs of about 99% for preferential matching, while the remaining 1% comes from the second step of this method (when it searches outside the treated unit's cluster to find a control match). It is important to notice that both the pure and preferential within-cluster matching show good performances also when the cluster-level variable Z is not a confounder ($\beta_Z = 0$). Therefore, these methods seem to be robust to overadjustment at the cluster-level in the presence of big clusters.

With respect to the intra-class correlation coefficients, we found that $ICC(T)$ increased with increasing values of β_Z , while $ICC(Y)$ was unaffected because β_Z only regulates the effect of Z in the true treatment Equation (7). An increased value of β_Z (and consequently of $ICC(T)$) indicates a stronger confounding effect of Z . The relative bias of methods W, FE and RE were rather insensitive to higher values of $ICC(T)$.

Finally, we notice that there was no substantial difference among the five considered PSM methods with respect to the root mean squared error and the average estimated standard error (SE). We can also notice that the average number of times a control units was used as a match tended to increase with the strength of the unobserved cluster-level confounder but it remained very small in all simulation conditions (maximum average value is 0.23). Therefore, the low number of unmatched units that we found for all methods with the exception of the within-cluster matching was due to the overall good overlap of the propensity score distribution across the two-treatment groups (Figure 1) and not to the replacement option.

As mentioned in the previous section, we considered different variations of the baseline simulation study. For these additional simulations, we report full results in tables similar to Table II in Section C of the Supporting Information for the sake of saving space. Here, we focus on comparing the relative bias across the different simulations graphically (Figure 2). The top left graph in Figure 2 also reports the relative bias from the previously commented baseline simulation for comparison.

First of all, we considered the same simulation as previously in the case the cluster-level confounder, Z , is known. In this situation, the researcher would include Z as a covariate in the propensity score models of approaches NV, PW and RE. The full results are shown in Table C1 in the Supporting Information. The top right graph in Figure 2 shows that when Z is observed, all methods perform very well in terms of bias reduction including the naive method that uses a standard logistic regression but now including the cluster-level covariate. The PW also benefits from including Z in the PSM. In fact, if matching was not possible within-cluster, including Z in the propensity score increases the chances to find a match in another cluster with similar values of the cluster-level confounder. The performance of this method improves especially for the highest values of β_Z . In fact, when Z is unobserved (top left of Figure 2), the relative bias of the preferential matching was 3.15% and 6.34% for the two highest values of β_Z (0.4 and 0.6, respectively). The bias reduces to 0.85% and 4.24%, respectively, when Z is observed. The relative bias of the RE approach is not substantially affected by observing Z . Also, the other quantities (number of unmatched treated units, number of controls replications, $RMSE$, etc.) do not change significantly.

As a second variation of the baseline simulation, we considered a simulation where the magnitude of the individual-level confounders in the true treatment assignment equation in a subset of 10 clusters was increased by 25%. Full results are reported in Table C2 of Section C of the Supporting Information. The bottom left graph in Figure 2 shows the relative bias of the different PSM approaches in this additional simulation.

The pattern across estimators remain similar to that observed in the baseline simulation (Table II or top left of Figure 2). However, higher relative biases are observed for all PSM approaches. This is reasonable as the propensity score models are now further misspecified. In fact, they assume homogeneous effects of individual-level confounders, while the effect of some of them varies across clusters. Nonetheless, it is reassuring that even in these scenarios, characterized by quite extreme heterogeneous effects for some

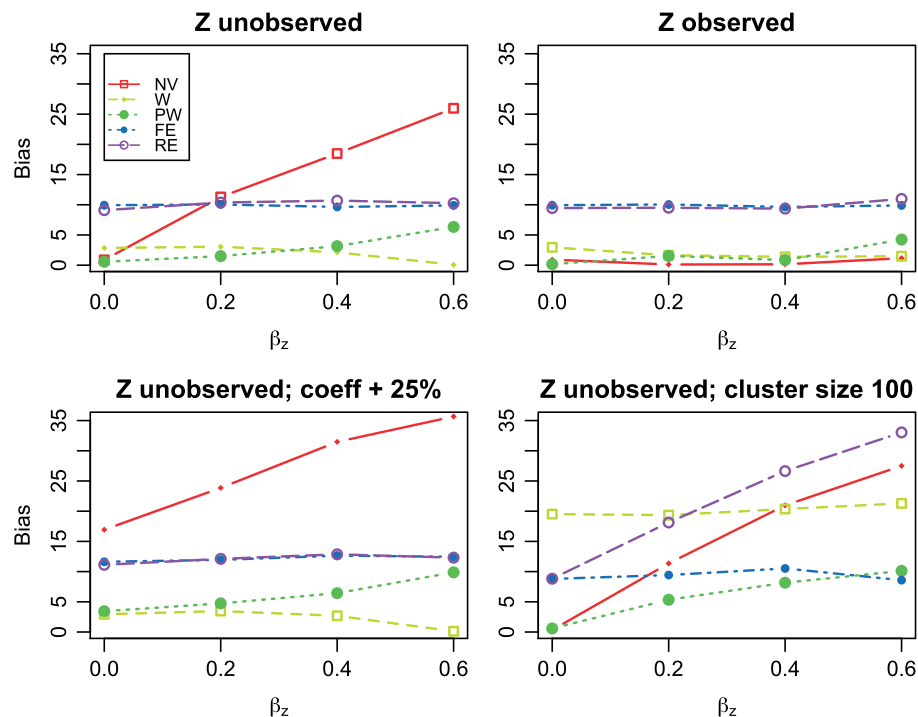


Figure 2. Comparison of percent relative bias for different strategies and simulation conditions.

individual-level confounders, all PSM methods that adjust for clustering (W, PW, FE and RE) maintain acceptable levels of relative biases. In particular, the pure within-cluster matching (method W) shows a low relative bias in all scenarios (always lower than 4%). The gap between PW and pure within-cluster (W) matching in terms of relative bias that we observed in the baseline simulation (top left of Figure 2) is even wider in the case where coefficients of individual-level variables were increased (bottom left). For the case where $\beta_Z = 0.6$, the gap between the relative bias of PW as compared with that of W increases from about 6 to about 10 percentage points. However, the relative bias of the PW remains acceptable and lower than that of the FE and RE approaches.

The number of unmatched treated units only slightly increased for the within-cluster matching (by about 2 units, on average) as compared with the baseline simulation (see Table C2 in the Supporting Information). This increased number of unmatched units was due to the fact that caliper was set using the propensity score model estimated on the pooled dataset assuming constant effects of individual-level confounders on the probability of treatment. In those clusters where the effect of individual-level confounders was stronger, the variance of propensity scores was higher than in the pooled sample, and it was more likely not to find a matched control within the caliper. For the other methods, the number of unmatched treated units remained very small (never higher than 0.5 units, on average). Also the average number of control replications for all PSM methods was small (never higher than 0.15 units, on average). The $RMSE_{montecarlo}$, as well as the average estimated standard error (SE), of the different PSM estimators was only marginally higher than in the baseline simulation.

As a third variation of the baseline simulation, we also assessed the performance of the various PSM estimators in the presence of smaller clusters. We replicated the baseline simulation for values of $\beta_Z = \{0, 0.2, 0.4, 0.6\}$ in the case of clusters' sizes equal to 100 units (full results in Table C3 of the Supporting Information). The bottom right graph in Figure 2 shows the relative bias of the different PSM approaches in this additional simulation. As expected, the relative bias of the within-cluster matching was much higher in this case as compared with the baseline simulation (top left). With small clusters (average size about 100 units), the bias of the within-cluster matching was about 20% for each of the values of β_Z , while in the baseline simulations (average cluster size equal to 738 units), its relative bias was always smaller than 3%. The higher bias of the within-cluster matching was due to the increased difficulty to find for each treated units a good control match within the same cluster. This is also shown by the substantively higher number of unmatched treated units (between 312 and 470 units in the different scenarios; see Table C3 in the Supporting Information) compared with the baseline simulation (between 52 and 61 units; see

Table II). On the other hand, the PW was able also in this case to preserve a good performance both in terms of low relative bias (always below 10%) and low number of unmatched treated units (always smaller than 0.64, on average). In the presence of small clusters, the consequence of a higher number of unmatched treated units for within-cluster matching was that for preferential matching the percentage of matched pairs that were found within clusters was reduced (from about 99% in Table II to about 92% in Table C3).

We also confirm the results in Arpino and Mealli [8] reporting high bias of the PSM based on a random-effects propensity score model in the presence of small clusters, especially when the effect of the unobserved cluster-level confounder is strong. The RE method was not able to achieve a good balance of Z : $ASB(Z)$ reached 51% when the effect of Z was the strongest (Table C3). This simulation with a lower average cluster size was the only case where we noticed a strong relationship between the relative bias of RE and $ICC(T)$: the higher was $ICC(T)$ the higher was the relative bias. The fixed-effects approach, instead, performed reasonably well in all cases.

To better appreciate how the bias of the different PSM approaches changes as the clusters' sizes change, we also considered additional simulations where the clusters' sizes were set to 50, 200 and 300 units (full results for the case $\beta_Z = 0.4$ are in Table C4 of the Supporting Information). In Figure 3, we show how the relative bias of the five PSM approaches we considered changed as the samples were varied. We include also results for an average cluster size of 738 units that corresponds to the baseline simulation. Results are for β_Z set to 0.4. Figure 3 shows that the pure within-cluster matching and the random-effect approach were the most sensitive methods to the variations in the clusters' sizes among the five methods considered. The relative bias of both the W and RE PSM estimators considerably increased as the average cluster size decreased. With an average cluster size of 50 units, both methods showed a relative bias higher than 25%. The within-cluster matching performed reasonably well already with an average cluster size of 300 units (relative bias = 12%). The PW, although showing better performances for bigger clusters, it always attained relative biases smaller than 10%, and it was the best performing method, apart in the last case with big clusters (average size = 738 units) where the bias of the pure and preferential matching were very similar.

A final simulation aimed at assessing the role of the presence of a positive within-cluster correlation (ICC) for covariates. In the presence of a positive ICC for a given individual-level covariate, X , (i.e. when the cluster means of X vary over clusters), propensity score methods that include X are expected to improve their relative performance because the cluster means of X can act as a proxy for the unmeasured cluster-level covariate, Z . The stronger the correlation between X and Z , the better will be the performance of PSM that adjust for X relative to the case where X and Z are not correlated. However, increasing levels of the $ICC(X)$ can have an effect on the ability of the different methods to balance X . It is difficult to say a-priori how the relative performance of the methods we studied would change in the case of a positive ICC for individual-level independent variables. To shed some light on this aspect, we added another small simulation experiment where we introduced a correlation between one of the independent variables, X , and Z and considered two levels of $ICC(X)$. In particular, we generated a new variable X' as $X' = X + \rho Z + \eta_j$, where $\eta_j \sim N(0, \sigma_\eta)$. We considered two levels of $\sigma_\eta = \{0.2, 0.6\}$ to generate two levels of $ICC(X') \approx \{10\%, 30\%\}$. ρ was set equal to 0.6 to generate a correlation between X' and

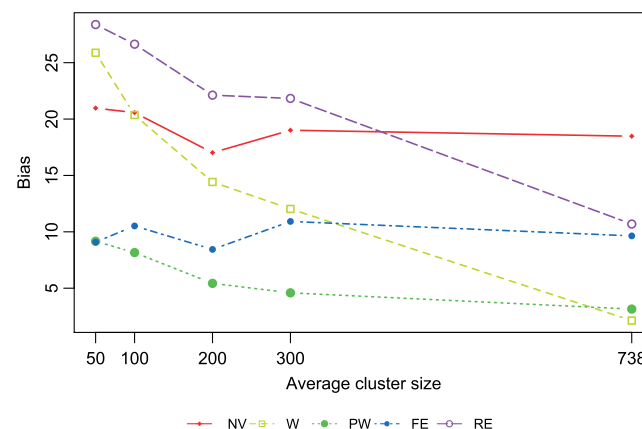


Figure 3. Comparison of percent relative bias for different strategies and average clusters' sample sizes.

Z of about 0.3. The new variable X' was used instead of the original variable X in the data generating Equations (7) and (8) (full results in Table C5 of the Supporting Information). We found that, as expected, all methods improved the balance of the unobserved cluster-level confounder Z . However, for the within-cluster and PW methods, the balance of the individual-level variables was worse than it was in the baseline simulation. This is due to a worse balance of the new generated variable X' . As a consequence, the relative bias for these two methods increased up to about 20% for PW in the case of $ICC(X') \approx 30\%$. On the contrary, the relative bias of RE and FE slightly decreased as a consequence of the better balance of Z . These results confirm the importance of considering the balance of each covariate in evaluating the quality of a PSM solution.

5. Empirical results

In this section, we report the results of the five PSM approaches applied to the motivating case study introduced in Section 2. The interest lies in estimating the effect of caesarean section on infants' Apgar score. In our empirical analyses, we adjusted for the following covariates: maternal age (five dummy variables: < 20 (reference), 20–24, 25–29, 30–35, >35 years), maternal education (three dummy variables: less than high school (reference), high school, graduate or higher education), infant birth weight (three dummy variables: < 2500, 2500–4000 (reference), > 4000 g), induction of labour ($yes = 1$, $no = 0$), gestational age (in weeks), and presence of a pregnancy-related pathology (binary variable set to 1 if one or more of the following diseases occurred during pregnancy: diabetes mellitus, eclampsia, hypertension, placenta previa). Table III reports the standardized mean difference for each covariate calculated on the raw data ('Crude') and then after the implementation of each of the five PSM method. The group of mothers that gave birth with a caesarean section (treated) and those who delivered naturally (control) were quite different with respect to several covariates. For example, the imbalance was relatively high for clinical indications of caesarean section, such as older age, lower gestational age and the presence of pathologies during the pregnancy. Overall, the average ASB was about 14%.

Each of the five PSM approaches considered substantially reduced the original imbalance. The naive PSM (NV) guaranteed the lowest average ASB. However, this method ignores the presence of possible unobserved cluster-level confounders and, as showed by the simulations, gives biased estimates when such confounders exist. Among the other methods, the fixed-effects and random-effects PSM gave slightly lower average ASB for the individual-level covariates than the other two methods (W and PW).

Variable	Strategy [†]					
	Crude	NV	W	PW	FE	RE
Maternal age (years)						
< 20	14.94	0.53	0.78	0.78	1.35	1.01
20–24	12.64	0.70	0.74	0.60	0.02	0.93
25–29	15.04	0.83	2.49	2.53	0.65	0.33
30–35	6.11	0.67	1.08	1.08	0.80	1.98
> 35	26.67	0.16	3.43	3.39	0.01	1.48
Maternal education						
Less than high school	2.53	1.68	2.63	2.91	3.57	2.88
High school	0.57	1.14	0.44	0.54	2.52	1.03
Graduate or more	2.80	0.32	2.61	2.77	1.69	0.99
Missing	0.06	0.46	0.57	0.68	0.43	1.94
Infant weight (grams)						
< 2500	21.50	0.42	4.39	4.30	0.74	0.89
2500–4000	23.90	3.00	5.52	5.32	0.08	0.10
>4000	9.14	1.34	3.00	2.79	0.10	1.20
Labour induction	5.04	3.68	2.00	2.07	3.29	2.29
Gestational age	32.44	2.15	2.46	2.71	2.13	2.70
Pathology during pregnancy	20.76	1.21	5.03	5.55	0.94	2.38
Average absolute standardised bias	13.94	1.21	2.48	2.54	1.28	1.48

[†] NV, naive; W, within-cluster; PW, preferential within-cluster; FE, fixed-effects; RE, random-effects.

Table IV. Results from crude and PSM estimates of the effect of caesarean section on the Apgar score.

Metrics	Strategy [†]					
	Crude	NV	W	PW	FE	RE
Number unmatched units	—	0	47	0	0	0
Average number of control replicates	—	0.08	0.16	0.17	0.10	0.10
Average ASB	14.80	1.21	2.48	2.53	1.28	1.48
$\widehat{ATT}(\%)$	5.75	2.80	2.62	2.25	2.68	2.07
$SE(\%)$	1.46	2.75	2.13	2.26	1.49	2.53

[†]NV, naive; W, within-cluster; PW, preferential within-cluster; FE, fixed-effects; RE, random-effects. ASB, absolute standardized bias; PSM, propensity score matching.

The number of unmatched treated units due to the caliper option was very small for the pure within-cluster matching (47 treated units), while for the other methods, all treated units could find a matched control unit within the imposed caliper. This is due to the overall good overlap in the distribution of propensity scores among treated and control units and to the large set of control units available with respect to the treated units. Table IV shows that, consistently with simulations results shown in the previous section, the average number of times a control units was used as a match is very low.

In Table IV, we show the *ATT* estimates calculated on the raw data and using the five PSM approaches. All methods consistently report a positive *ATT* estimate indicating that caesarean section increases the risk of low Apgar scores. The unadjusted difference in the prevalence of low Apgar scores between the treatment and control groups seems to be substantively overestimated. In fact, all five PSM approaches considered gave considerably lower *ATT* estimates. Interestingly, all methods that adjust for clustering (W, PW, FE and RE) gave lower estimates than the naive PSM. This may indicate a possible overestimation of the effect of caesarean section when hospital confounding effects are not taken into account. However, we notice that for all PSM estimators the standard error of the *ATT* is high compared to the estimated *ATT* and tests of hypothesis would not reject the null hypothesis of zero effect at the standard levels of significance.

6. Summary and concluding remarks

Propensity score methods have been typically considered in the context of unstructured data. In this paper, we considered PSM for clustered data. Inspired by a real case study regarding the estimation of the effect of caesarean section on the Apgar score with data on mothers nested within hospitals, we evaluated via Monte Carlo simulations different strategies to deal with the presence of unobserved confounders at the cluster-level. We assessed two different types of approaches. We considered estimation of the propensity score model using fixed-effects or random-effects logistic regressions. Additionally, we considered matching on a single-level propensity score within clusters. We also proposed a less extreme variant of the latter approach, which we labelled as preferential within-cluster matching. This method first searches for control units to be matched to treated units within the same cluster. If matching is not possible within-cluster, then the algorithm searches in other clusters.

Differently, from Arpino and Mealli [8], our baseline simulation study considered a strongly unbalanced structure where some clusters were big enough to justify within-cluster matching, while others had small sample sizes. We confirmed on this type of structure the ability of fixed-effects and random-effects propensity score models to reduce bias because of omitted cluster-level confounders when used to implement matching. However, we found that the pure and PW approaches performed better. In particular, PW was the best method when the strength of the omitted confounder was low or medium, while within-cluster matching had the best performance in the case of a strong omitted confounder. Within-cluster matching demonstrated to be robust to the introduction of varying coefficients of individual-level confounders across clusters but did not perform well when all clusters were small. In this case, PW worked much better.

An important aspect to consider when implementing PSM is that a high proportion of unmatched units may imply an implicit change in the estimand [12]. As expected, our simulations showed that imposing a pure within-cluster matching may produce a high proportion of unmatched units, especially when clusters

sizes are small. Moreover, we noticed (analyses not reported but available upon request) that the number of unmatched units was not proportional to the cluster size, because the distribution of covariates may vary across hospitals. Therefore, within-cluster matching can imply a change in the estimand not only because some treated units can remain unmatched but also because the data structure can be altered by changing the relative sizes of the different clusters. The PW offers a solution to this problem, and its performance was good in all scenarios both in terms of relative bias and number of unmatched units. Optimal matching is an alternative approach to PSM that has the advantage of avoiding unmatched units. Its extension to multilevel settings is under-studied (see [40] for an exception). Future studies can compare the performance of the methods we analysed in this paper and new optimal matching algorithms for multilevel data.

In our simulations, we assumed that the treatment assignment mechanisms followed a parametric logistic model. We also used a simple data generating process where covariates were included only additively in the linear predictor. However, the true propensity score model may take a complicated form, and the propensity score model may be misspecified. Machine learning techniques have been proposed as a more flexible way of estimating propensity scores than parametric models [41]. The application of these methods in the context of clustered data structures is an interesting avenue for future research.

The propensity score distribution showed a good overlap between treated and control group in the simulation experiments we implemented. Future investigations are needed to assess if results we found, and especially the good performance of the PW approach, apply also in cases where overlap is poor.

We included all (potential) confounders in the estimation of the propensity score model. There is a lack of consensus in the causal inference literature as to which variables to include in the propensity score model. However, most authors advise to include prognostically important covariates (those related to the outcome) and confounding covariates (those related to both the treatment and the outcome) in the propensity score model and to avoid including those variables that affect only the treatment (e.g. [42]). Both for unstructured and clustered data the guiding principle for selecting covariates should be theory and past evidence. Covariates selection for PSM with clustered data has not been considered so far, and it is another topic to be analysed in future studies.

Summarizing our study confirmed the importance of considering clustering in the estimation of causal effects via PSM. Our simulations offer several insights to applied researchers for the choice of the PSM approach in the presence of clustered data. The choice among the methods should mainly depend on the data structure. We showed that when all or at least the majority of clusters' sample sizes are big, an effective approach consists in implementing the matching within clusters. When cluster sizes are small, within-cluster matching may imply losing a high proportion of units and may be seriously biased. Our simulations suggest that already with an average cluster size of about 100 units the bias of within-cluster matching can be rather high. With smaller clusters of size 50, our results were even more negative about the use of pure within-cluster matching. Our proposal of a preferential within-cluster matching is a better alternative in these cases. The preferential cluster matching approach, combining the advantages of within-cluster and between-cluster matching, showed a relatively good performance both in the presence of big and small clusters, and it was often the best method. Also fixed-effects PSM worked well with small clusters confirming previous results by Arpino and Mealli [8]. The performance of the pure within-cluster matching started to be acceptable with clusters of size 300.

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